Last time: Universal vs. existence

Conditional $p \to q$

direct: $p \to \cdots \to q$

Lesson 7: Biconditional; proof by contrapositive

$p \leftrightarrow q$

$= (p \to q) \wedge (q \to p)$

Biconditional

$p \to q = \sim q \to \sim p$

contrapositive

$\sim q \to \cdots \to \sim p$

Prop 1 $\forall x \in \mathbb{Z}$, x is odd $\Leftrightarrow$ $x-1$ is even.

iff

Proof ($\Rightarrow$) [x is odd $\to$ $x-1$ is even]

x is odd $\to$ $x = 2n+1$, $n \in \mathbb{Z}$

$\to$ $x-1 = 2n$

even

($\Leftarrow$) [$x-1$ is even $\to$ x is odd]

$x-1$ is even $\to$ $x-1 = 2n$, $n \in \mathbb{Z}$

$\to$ $x = 2n+1$

odd

Proof by contrapositive

Prop 2 $\forall x \in \mathbb{Z}$, if x^2 is even, then x is even

hard to prove from x^2 even to x even

<u>Proof</u> We will prove the <u>contrapositive</u>:

If x is odd (not even), then x^2 is odd.

x is odd $\to x = 2n+1$

$\to x^2 = (2n+1)^2 = 4n^2+4n+1$

$= 2\underbrace{(2n^2+2n)}_{\in \mathbb{Z}}+1$

$\to \quad x^2$ odd

$2 \mid x^2 \to 2 \mid x$

(even) (even)

General version:

$\boxed{p \mid x^2 \to p \mid x}$ (prove in lesson 10)

prime

<u>Prop 3</u> $\forall a, b \in \mathbb{R}$, if $ab = 1$, then $a \le 1$ or $b \le 1$.

<u>Direct proof</u>

If $a \le 1$, then we are done.

If $a \not\le 1$, then $a > 1$, and $b = \frac{ab}{a} = \frac{1}{a}$

$b = \frac{1}{a} < 1 \Rightarrow b \le 1$

$\text{P} \to p \text{ or } q$

Proof by contrapositive

Contrapositive: If $a>1$ and $b>1$, then $ab \neq 1$.

$$ab > a(1) > 1$$

Prop 4 The product of an even integer and any integer is even.

Proof

WLOG

$$\underset{\text{even}}{x}y = (2n)y = 2\underbrace{(ny)}_{\text{even}}$$

Prop 5 If the product of 2 integers is odd, then they are not consecutive.

Proof We will prove the contrapositive:

If 2 integers are consecutive, then their product is even.

Let x and x+1 be consecutive integers

If x is even, then by Prop 4, $x(x+1)$ is even.

If x is odd, then $x=2n+1$ and $x+1=2n+2$

$$=2(n+1)$$

So $x+1$ is even, and $x(x+1)$ is even by Prop 4.

Prop 6 $\forall x \in \mathbb{Z}$, x is odd iff x^2-1 is even.

$$p \leftrightarrow q = (p \to q) \wedge \underbrace{(q \to p)}_{\text{hard}}$$

$\sim p \to \sim q$

(contrapositive of $q \to p$)

$$p \leftrightarrow q = (p \to q) \wedge (\sim p \to \sim q)$$

inverse of $p \to q$

($\Rightarrow$) x is odd $\to$ $x = 2n+1$

$$x^2-1=(2n+1)^2-1$$
$$=4n^2+4n$$
$$=2(2n^2+2n)$$

x^2-1 is even

($\Leftarrow$) contrapositive

x is <u>not</u> odd $\rightarrow$ x is even

$\rightarrow x = 2n, \ n \in \mathbb{Z}$

$\rightarrow x^2 - 1 = (2n)^2 - 1$

$= 2\underbrace{(2n^2 - 1)} + 1$

$\rightarrow x^2 - 1$ is odd

(not even)